

Credential Attack Simulation and Log-Based Detection in a Controlled Environment

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Kali Linux · Windows Server 2022 · Active Directory · Hydra · Nmap · Event Viewer · Wazuh

Project at a glance: Completed as a guided lab exercise based on a Udemy course recommended by a colleague: "Active Directory Lab with Nmap and Hydra: Build a Windows Server 2022 and Linux Lab for CySA+ and Sec+." The course covers building a full Windows Server 2022 and Linux lab from scratch using free tools, then going hands-on with Nmap, Hydra, Active Directory enumeration, Wazuh, and EICAR. I worked through the core attack and detection sections, adapting where my environment differed from the course setup: VMware Fusion on macOS instead of VirtualBox, hardware constraints that prevented a full Wazuh deployment, and protocol-level blocks that changed how Hydra behaved against RDP and SMB. The EICAR malware simulation section was not completed in this run. The result was a functional lab that produced real authentication failure logs and grounded how brute-force activity appears from a defender's perspective.

Executive Summary

A colleague recommended the Udemy course "Active Directory Lab with Nmap and Hydra: Build a Windows Server 2022 and Linux Lab for CySA+ and Sec+," and I decided to work through it as a hands-on way to build out my home lab experience. The course is built around free tools and walks you through building a complete Windows Server 2022 plus Linux environment from scratch, configuring a Domain Controller, adding a Linux attacker machine, and then going hands-on with Nmap for network discovery, Hydra for brute-force attacks, Active Directory enumeration basics, Wazuh for detection and alerting, and EICAR for safe malware simulation.

My setup differed from the course in a few meaningful ways. The course assumes VirtualBox as the free virtualization platform, but I was running VMware Fusion on macOS, which required adjusting the network adapter configuration to get the two VMs communicating. The Wazuh deployment ran into hardware limitations because my Ubuntu VM did not meet the 4 GB RAM minimum, so after two attempts I pivoted to native Windows Event Viewer for the detection work. The EICAR malware simulation section was not completed in this run.

Despite those gaps, I completed the core attack and detection objectives: confirmed network connectivity, mapped the attack surface with Nmap, executed brute-force attempts against RDP and SMB using Hydra, and confirmed that the activity generated 13 Event ID 4625 entries in the Windows Security log within a six-minute window. The places where my environment pushed back against the course instructions ended up being the most instructive parts of the whole exercise.

Project Objective

The goal was to work through a structured lab exercise that combined offensive tooling with defensive visibility, using a published course as the scaffolding and my own environment as the testing ground. Beyond following the steps, I wanted to understand how failed authentication attempts look from a defender's perspective and whether native Windows logging alone could

provide enough signal to detect brute-force activity without a SIEM in place.

Scenario

The course is framed around giving beginners real, practical skills rather than theory or memorization. It positions each tool section as something security analysts use every day, so the expectation is that you understand not just how to run the tools but why they matter. I approached the exercise with that framing in mind: setting up the environment from scratch, working through tool failures as I would on the job, and making deliberate decisions about when to adapt the course steps and when to pivot to an alternative approach entirely. The goal was to get to the same destination the course points toward, using whatever path my environment allowed.

Environment and Architecture

The lab consisted of a Kali Linux attacker machine and a Windows Server 2022 target with Active Directory configured, both running as virtual machines. The course uses VirtualBox; I used VMware Fusion on macOS, which required adjusting the network adapter configuration to establish host-only connectivity between the two machines. A third Ubuntu VM was brought up specifically to attempt the Wazuh SIEM deployment.

Component	Role	Key Configuration	Observed Data
Kali Linux	Attacker system	Hydra v9.6, Nmap 7.99	Attack execution and output
Windows Server 2022	Target system	Active Directory (lab.local)	Authentication failure logs
Windows Event Viewer	Log analysis tool	Security log, Event ID 4625	13 failed logon events
Wazuh 4.7.5 (Ubuntu VM)	SIEM deployment attempt	Resource-limited install (under 4 GB RAM)	Indexer service failure

Methodology

The course follows a progression from environment setup through to detection analysis. I followed the same overall sequence, noting where I had to deviate:

1. Verified network connectivity between Kali and Windows Server using ICMP ping
2. Ran an Nmap service version scan to identify exposed ports and services
3. Confirmed SMB and RPC-related services (ports 135, 139, 445) as the attack surface
4. Created a test domain user account (student1) in Active Directory
5. Prepared username and password wordlists on Kali
6. Ran Hydra against RDP (port 3389) as the course instructs, which returned freerdp connection failures on my setup
7. Pivoted to SMB (port 445), which returned an invalid reply from the target
8. Moved to the Windows Server VM and opened Event Viewer to inspect the Security log
9. Filtered for Event ID 4625 to identify and count failed logon attempts generated by the attack

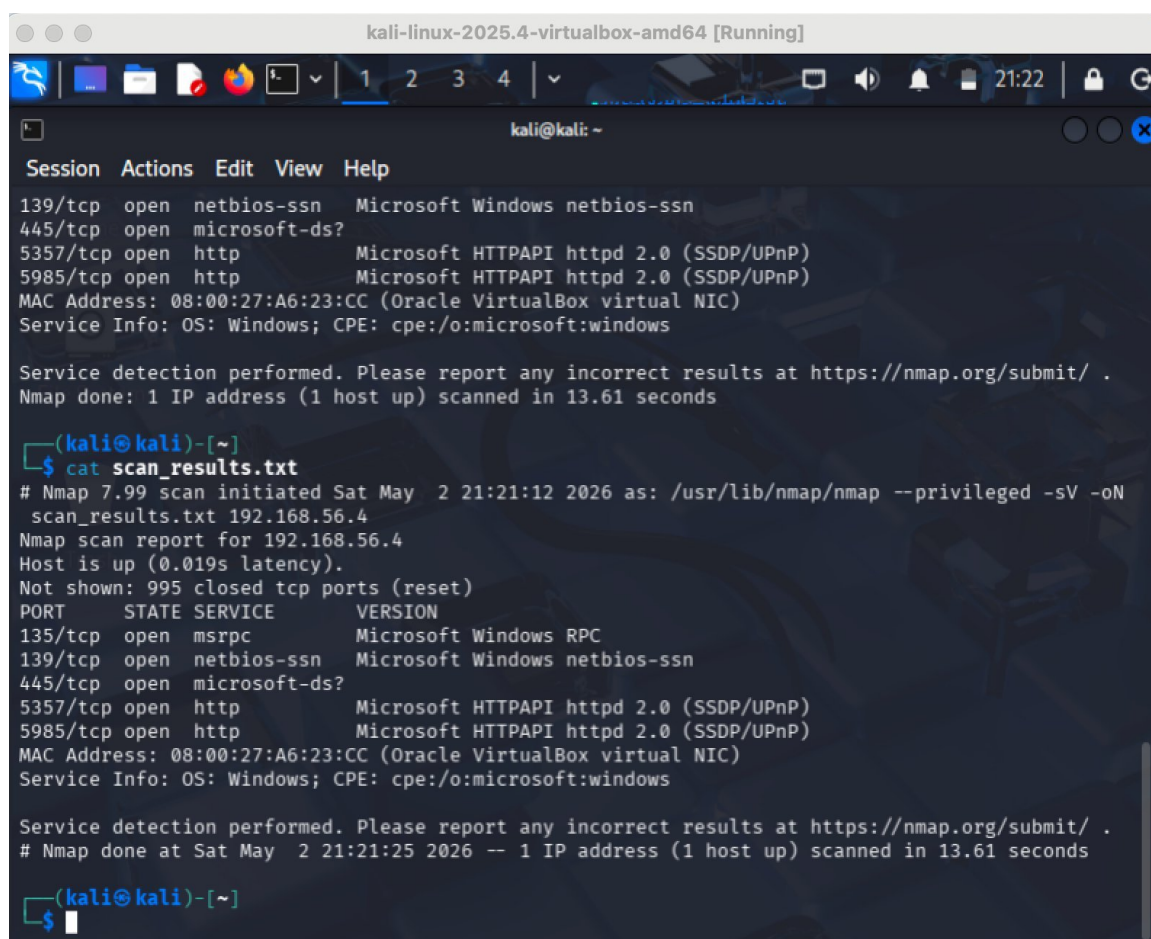
10. Attempted Wazuh 4.7.5 installation on a separate Ubuntu VM following the course instructions
11. Troubleshot the hardware requirements error, re-attempted with the -i flag, documented the indexer failure, and pivoted to native log analysis

Key Activities and Findings

1. Network Discovery and Service Identification

Before running any attacks, I confirmed reachability between the two VMs using ICMP ping, then ran an Nmap service version scan against 192.168.56.4. The scan finished in 13.61 seconds and came back with five open ports: 135 (MSRPC), 139 (NetBIOS-SSN), 445 (Microsoft-DS), 5357 and 5985 (both HTTPAPI). I saved the output to scan_results.txt. The presence of SMB on 445 and NetBIOS on 139 confirmed there were authentication surfaces to target in the next phase.

Security relevance: Reconnaissance before an attack is the same process a defender uses to understand exposure. Knowing which ports are open and what services are running is what determines where authentication telemetry is going to appear.



```
kali-linux-2025.4-virtualbox-amd64 [Running]
kali@kali: ~
Session Actions Edit View Help
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
445/tcp open microsoft-ds?
5357/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5985/tcp open http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 08:00:27:A6:23:CC (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.61 seconds

(kali@kali)-[~]
$ cat scan_results.txt
# Nmap 7.99 scan initiated Sat May 2 21:21:12 2026 as: /usr/lib/nmap/nmap --privileged -sV -oN scan_results.txt 192.168.56.4
Nmap scan report for 192.168.56.4
Host is up (0.019s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 08:00:27:A6:23:CC (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Sat May 2 21:21:25 2026 -- 1 IP address (1 host up) scanned in 13.61 seconds

(kali@kali)-[~]
$
```

Figure 1. Nmap service version scan confirming open ports and exposed services on the Windows Server 2022 target.

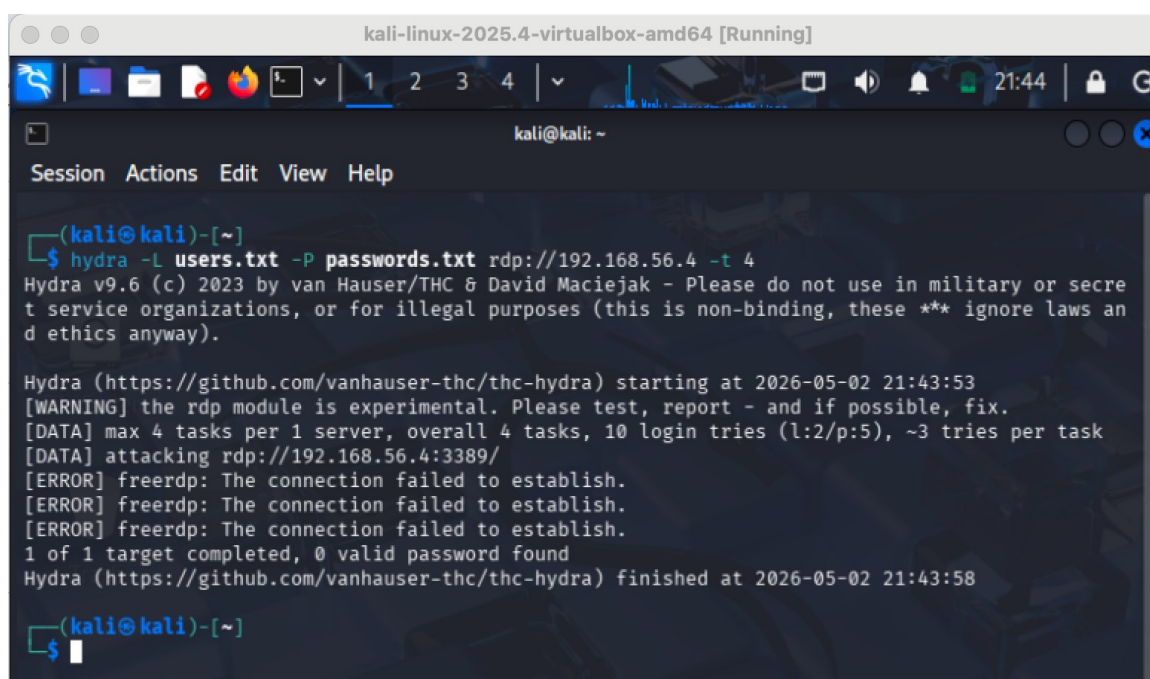
2. Credential Attack Simulation Using Hydra

The course walks through running Hydra against RDP, so I started there using prepared username and password wordlists with four parallel tasks across 10 login attempts. Hydra returned freerdp

connection failures on all three threads and finished in about five seconds with zero valid credentials. This is most likely because Network Level Authentication was blocking the connection before any credentials could even be evaluated, which Hydra's experimental RDP module cannot handle.

I pivoted to SMB as an alternative, which the course also covers. Hydra reduced itself to a single task automatically because SMB does not support parallel connections, then returned an invalid reply from the target after running through all 10 attempts. Neither vector got through, but that was fine because the point of the exercise was not to succeed at the attack. It was to see what the Windows Security log recorded as a result of the attempts.

Security relevance: The connection failures at the protocol level did not prevent authentication failure events from appearing in the Windows Security log. Detection happened regardless of whether the attack succeeded, which is exactly the lesson the course is designed to reinforce.

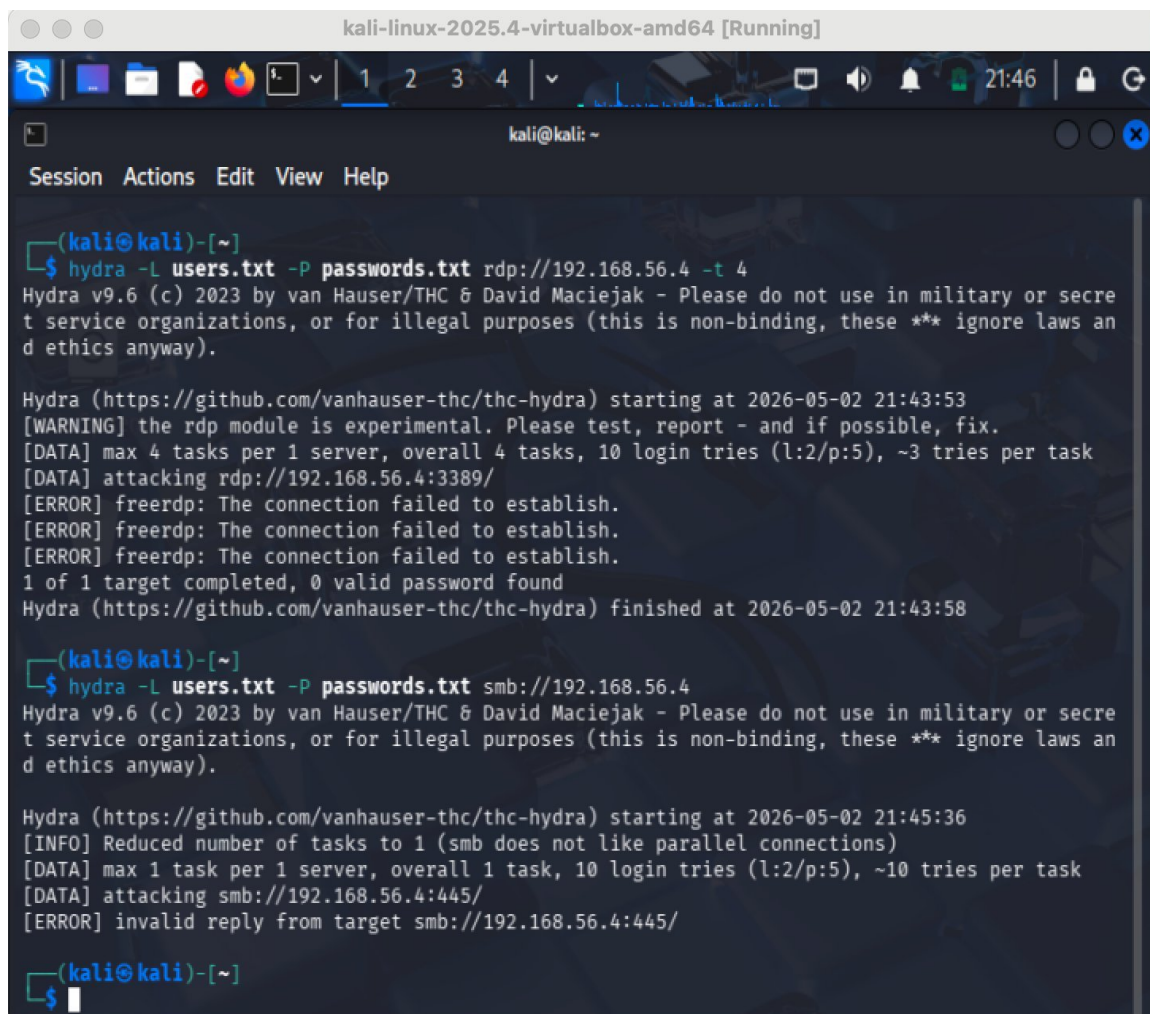


```
kali-linux-2025.4-virtualbox-amd64 [Running]
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ hydra -L users.txt -P passwords.txt rdp://192.168.56.4 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-02 21:43:53
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 10 login tries (l:2/p:5), ~3 tries per task
[DATA] attacking rdp://192.168.56.4:3389/
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-02 21:43:58

(kali@kali)-[~]
$
```

Figure 2. Hydra brute-force attempt against RDP showing freerdp connection failures and zero valid credentials found.



```
kali-linux-2025.4-virtualbox-amd64 [Running]
kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ hydra -L users.txt -P passwords.txt rdp://192.168.56.4 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-02 21:43:53
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 10 login tries (l:2/p:5), ~3 tries per task
[DATA] attacking rdp://192.168.56.4:3389/
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-02 21:43:58

(kali@kali)-[~]
$ hydra -L users.txt -P passwords.txt smb://192.168.56.4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-02 21:45:36
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[DATA] max 1 task per 1 server, overall 1 task, 10 login tries (l:2/p:5), ~10 tries per task
[DATA] attacking smb://192.168.56.4:445/
[ERROR] invalid reply from target smb://192.168.56.4:445/

(kali@kali)-[~]
$
```

Figure 3. Hydra against SMB returning an invalid reply, confirming both protocols were blocked at the connection level.

3. Log-Based Detection of Attack Activity

After running the attacks, I moved to the Windows Server VM and opened Event Viewer. Filtering the Security log for Event ID 4625 returned 13 entries between 6:38 PM and 6:44 PM on May 2, which matched the Hydra execution window exactly. Each entry showed the targeted account name, logon type, failure reason, and timestamp. What stood out was the clustering: multiple failures within the same second, which is not something a human login attempt would produce. That pattern alone is enough to flag automated activity in a real environment.

Security relevance: This was the most useful part of the whole lab for me. Even with the Wazuh SIEM out of the picture, the native Windows Security log gave a clear picture of what happened, when it happened, and what account was targeted. That is the core of what Event ID 4625 is for.

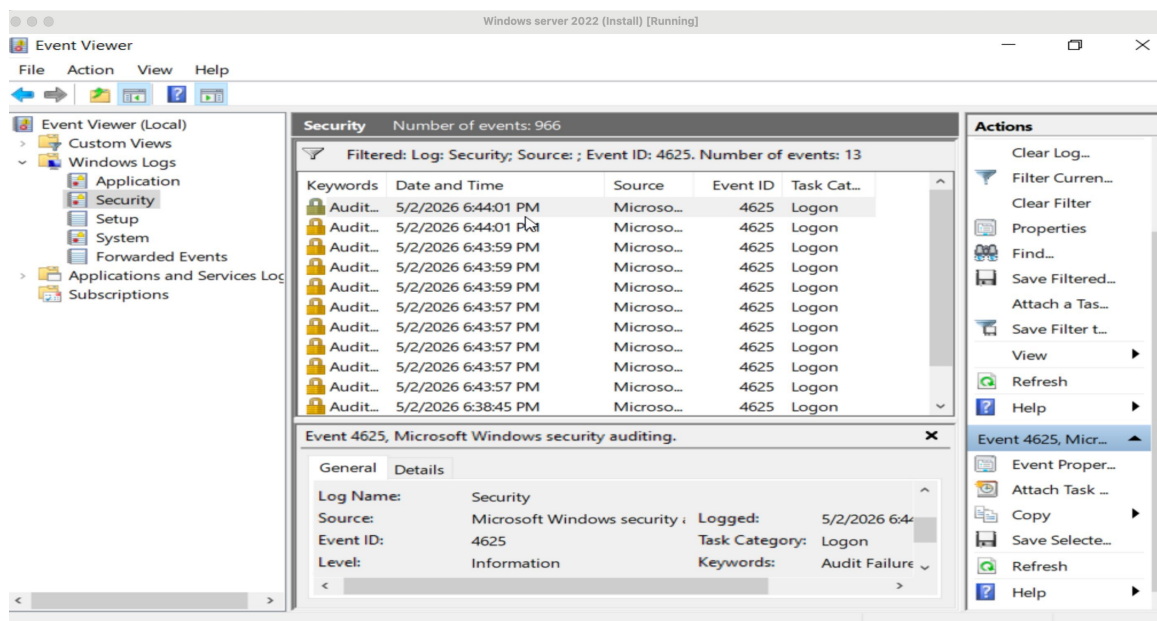


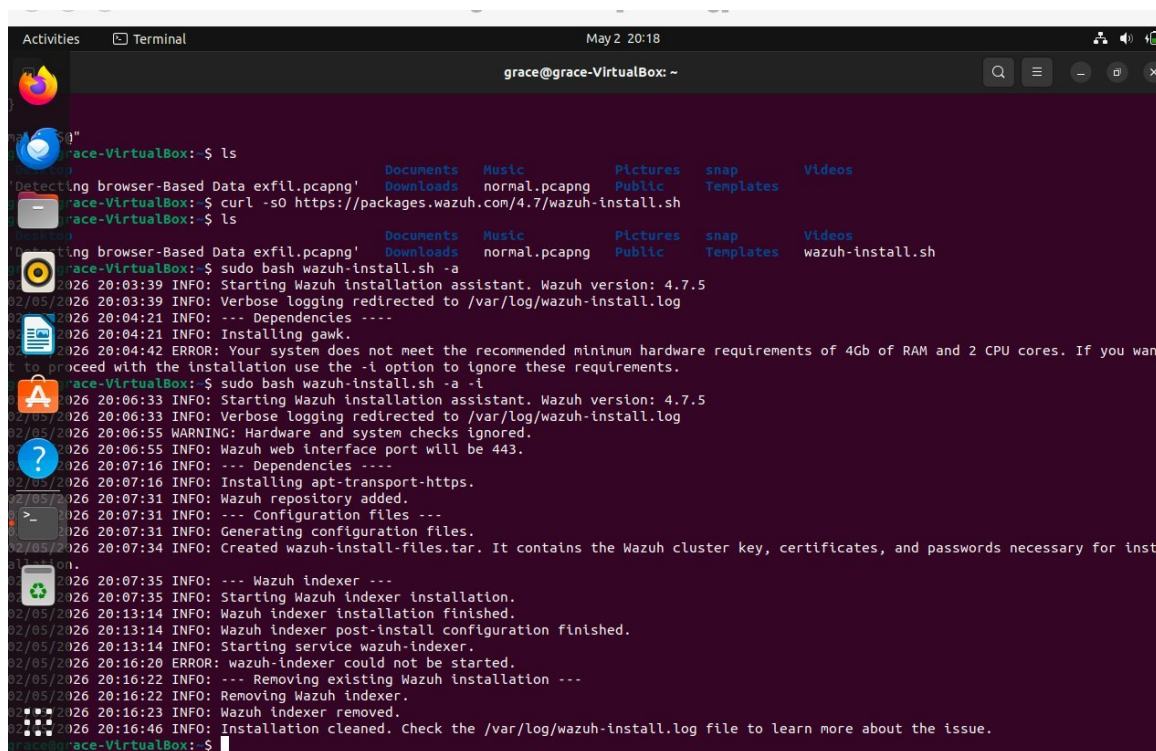
Figure 4. Windows Event Viewer Security log filtered by Event ID 4625, showing 13 failed logon events within the Hydra attack window.

4. Wazuh SIEM Deployment Attempt

The course includes a section on deploying Wazuh for centralized log monitoring and alerting, which I attempted on a separate Ubuntu VM (log01-ubuntu). The first run of the install script failed almost immediately with a hardware requirements error: my system did not meet the minimum of 4 GB RAM and 2 CPU cores.

I tried again using the `-i` flag to bypass the hardware check, which the installer itself suggests as an option. That run got much further, completing the dependency installation, repository configuration, certificate generation, and the indexer installation before failing when the wazuh-indexer service could not start. The installer cleaned itself up automatically and pointed me to the log file for the root cause. At that point I made the call to move on and complete the detection objective using native Event Viewer instead, which was available and working regardless of the Wazuh outcome.

Security relevance: This kind of constraint shows up in real security work all the time. Tools have minimum requirements, environments do not always cooperate, and the ability to pivot to what is available rather than abandoning the objective is what matters. The detection still happened.



```
grace@grace-VirtualBox: ~  
$ ls  
Documents Music Pictures snap Templates Videos  
Downloads normal.pcapng Public Templates wazuh-install.sh  
$ sudo bash wazuh-install.sh -a  
2026 20:03:39 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5  
2026 20:03:39 INFO: Verbose logging redirected to /var/log/wazuh-install.log  
2026 20:04:21 INFO: --- Dependencies ---  
2026 20:04:42 ERROR: Your system does not meet the recommended minimum hardware requirements of 4Gb of RAM and 2 CPU cores. If you want to proceed with the installation use the -i option to ignore these requirements.  
$ sudo bash wazuh-install.sh -a -i  
2026 20:06:33 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5  
2026 20:06:33 INFO: Verbose logging redirected to /var/log/wazuh-install.log  
2026 20:06:55 WARNING: Hardware and system checks ignored.  
2026 20:06:55 INFO: Wazuh web interface port will be 443.  
2026 20:07:16 INFO: --- Dependencies ---  
2026 20:07:16 INFO: Installing apt-transport-https.  
2026 20:07:31 INFO: Wazuh repository added.  
2026 20:07:31 INFO: --- Configuration files ---  
2026 20:07:31 INFO: Generating configuration files.  
2026 20:07:34 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.  
2026 20:07:35 INFO: --- Wazuh indexer ---  
2026 20:07:35 INFO: Starting Wazuh Indexer installation.  
2026 20:13:14 INFO: Wazuh indexer installation finished.  
2026 20:13:14 INFO: Wazuh indexer post-install configuration finished.  
2026 20:13:14 INFO: Starting service wazuh-indexer.  
2026 20:16:20 ERROR: wazuh-indexer could not be started.  
2026 20:16:22 INFO: --- Removing existing Wazuh installation ---  
2026 20:16:22 INFO: Removing Wazuh Indexer.  
2026 20:16:23 INFO: Wazuh Indexer removed.  
2026 20:16:46 INFO: Installation cleaned. Check the /var/log/wazuh-install.log file to learn more about the issue.  
$
```

Figure 5. Wazuh 4.7.5 installation output showing the hardware requirements error, the -i flag bypass attempt, and the wazuh-indexer service failure.

Challenges and How I Adapted

The biggest difference between my run through this lab and what the course shows was the Hydra behavior. The course demonstrates Hydra successfully generating connection attempts that show up cleanly in Event Viewer. In my environment, both RDP and SMB returned error messages at the connection level rather than authentication failures. I spent time reading through the Hydra output carefully and working out that NLA was the likely culprit for RDP, and that SMB's dialect negotiation was probably causing the invalid reply. Once I understood those as protocol-level rejections rather than tool failures, I checked Event Viewer and confirmed the 4625 entries were there anyway, which resolved the uncertainty.

The VMware Fusion versus VirtualBox difference also required some adjustment at the network configuration stage. The host-only adapter setup works differently between the two, so getting the Kali and Windows Server VMs to see each other took some troubleshooting before the ping test came back clean.

The Wazuh failure was straightforward to diagnose once I read the installer output carefully. The two-attempt approach (first run to surface the error, second run with -i to see how far it would get) was deliberate. When the indexer still could not start on the second attempt, I had enough information to understand that the hardware floor was a genuine blocker rather than a configuration issue I could work around.

Evidence Summary

- Network connectivity confirmed between Kali and Windows Server 2022 via ICMP ping
- Five open ports identified including SMB (445) and NetBIOS (139) as authentication surfaces

- Hydra brute-force executed against both RDP and SMB with zero valid credentials, both blocked at the protocol level
- 13 Event ID 4625 entries recorded in the Windows Security log within a six-minute window matching the attack timeframe
- Temporal clustering of failures per second confirmed as inconsistent with human login behavior
- Wazuh deployment attempted twice, hardware bypass attempted, indexer failure documented, and native detection confirmed as the fallback

What I Learned

Working through a structured course in your own environment instead of following it step for step is genuinely different. The course gives you the logic and the sequence, but your environment pushes back in ways the course cannot anticipate. That pushback is where the learning actually happens. I came out of this with a much clearer understanding of how NLA affects brute-force tooling, what Event ID 4625 actually contains and why it matters, and how to read an installer failure log well enough to know whether a problem is fixable or not.

The Wazuh section of the course is something I want to revisit once I have a machine with adequate resources. Getting the full SIEM layer working and seeing how it correlates with the same 4625 events I identified manually would close out the picture that this lab started.

Why This Project Matters

Hands-on labs like this one are how I close the gap between what I see in tools at work and what I understand about why those tools behave the way they do. Following a published course gives me a structured starting point, but running it myself in my own setup forces me to think through the decisions rather than just observe them. That combination of guided structure and independent troubleshooting is what makes this kind of self-directed learning worth documenting.

Brute-force against Active Directory is one of the most common initial access techniques in real enterprise environments, and being able to explain how it appears in logs and what behavioral markers distinguish it from normal failed logins is directly applicable to the SOC work I am doing now.

Next Steps

- Revisit the Wazuh deployment on a machine that meets the minimum hardware requirements and complete the SIEM integration section of the course
- Complete the EICAR malware simulation section that was skipped in this run, and observe how Wazuh surfaces it alongside the brute-force events
- Try the same Hydra exercise targeting a domain account where NLA is disabled to see the difference in both tool output and log behavior
- Expand the wordlists and test password spraying as a lower-noise alternative to traditional brute-force
- Map the observed behavior to MITRE ATT&CK T1110 (Brute Force) and T1110.003 (Password Spraying) and document the detection gaps
- Correlate the Nmap scan results and Hydra attempt timestamps in a single timeline to practice the kind of log correlation done in the DU SOC